

## Research paper

# Understanding structural-functional connectivity coupling in patients with major depressive disorder: A white matter perspective

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## ARTICLE INFO

## Keywords:

Major depressive disorder

White matter

Structural-functional connectivity

Coupling

Subgroups

## ABSTRACT

**Purpose:** To elucidate the structural-functional connectivity (SC-FC) coupling in white matter (WM) tracts in patients with major depressive disorder (MDD).

**Methods:** A total of 178 individuals diagnosed with MDD and 173 healthy controls (HCs) were recruited for this study. The Euclidean distance was calculated to assess SC-FC coupling. The primary analyses focused on investigating alterations in SC-FC coupling in WM tracts of individuals with MDD. Additionally, we explored the association between coupling and clinical symptoms. Secondary analyses examined differences among three subgroups of MDD: those with suicidal ideation (SI), those with a history of suicidal attempts (SA), and those non-suicidal (NS).

**Results:** The study revealed increased SC-FC coupling mainly in the middle cerebellar peduncle and bilateral corticospinal tract ( $P_{FDR} < 0.05$ ) in patients with MDD compared with HCs. Additionally, right cerebral peduncle coupling strength exhibited a significant positive correlation with Hamilton Anxiety Scale scores ( $r = 0.269$ ,  $P_{FDR} = 0.041$ ), while right cingulum (hippocampus) coupling strength showed a significant negative correlation with Nurses' Global Assessment of Suicide Risk scores ( $r = -0.159$ ,  $P_{FDR} = 0.036$ ). An increase in left anterior limb of internal capsule ( $P_{Bonferroni} < 0.01$ ) and left corticospinal tract ( $P_{Bonferroni} < 0.05$ ) coupling has been observed in MDD with SI. Additionally, a decrease in right posterior limb of internal capsule coupling has been found in MDD with SA ( $P_{Bonferroni} < 0.05$ ).

**Conclusions:** This study emphasizes the variations in SC-FC coupling in WM tracts in individuals with MDD and its subgroups, highlighting the crucial role of WM networks in the pathophysiology of MDD.

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## 1. Introduction

Major depressive disorder (MDD) is common, frequently recurring, and associated with an increased risk of suicide (Marwaha et al., 2023). Common manifestations of MDD include depressed mood, reduced interest or pleasure in activities, insomnia, fatigue, feelings of worthlessness and reduced ability to think or focus. It is the leading cause of mental-health-related disability worldwide (Herrman et al., 2019). However, at present, the etiology of MDD is still unknown.

Previous evidence has shown that MDD is the most common mental health disorder associated with disconnectivity in the brain (Anon, 2023). Research on brain networks is crucial for understanding the pathophysiological mechanisms underlying MDD (Tura and Goya-Maldonado, 2023). Most studies on the neural basis of MDD focus on the structural connectivity (SC) or functional connectivity (FC) of the gray matter (Gruber et al., 2023; Kaiser et al., 2015; Liao et al., 2024; Xia et al., 2022). However, white matter (WM), which is composed of axons connecting different nearby areas, plays a crucial role in complex cognitive activities (Stiles and Jernigan, 2010). Therefore, it is essential to investigate the role of WM in both its structure and function, as well as its potential involvement in MDD.

Microstructure damage in WM has been observed through diffusion tensor imaging (DTI), suggesting that alterations in WM may play a pivotal role in the etiology of MDD (Wu et al., 2023). Specifically, patients with MDD showed reduced fractional anisotropy (FA) in the genu and body of the corpus callosum, coronal radiation, and thalamic radiation. This suggests impaired microstructural integrity and lower myelination (Hu et al., 2024; Meinert et al., 2022; Wu et al., 2023). Furthermore, abnormal communication in brain SC has been observed in patients with MDD, which may mediate the association of cognitive deficits and depression severity (Gruber et al., 2023; Lemke et al., 2022; Zhang et al., 2020). Additionally, WM exhibits blood oxygenation level-dependent (BOLD) signals, which reflect neural activities both at rest and under functional demands (Ding et al., 2018). Recent evidence has shown that BOLD FC is organized into distributed networks in WM tracts (Wang et al., 2022), with an underlying electrophysiological and structural basis (Huang et al., 2023). Disruptions in the topological organization of WM functional connectomes have been observed in patients with MDD (Li et al., 2020). Additionally, FC in WM tracts has been significantly correlated with depressive severity and the improvement of symptoms induced by antidepressant treatment (Zhang et al., 2021). This evidence offers a novel perspective for investigating disturbances of structure and function in WM tracts in patients with MDD.

Relating the brain's SC to its FC is a fundamental goal in neuroscience (Wang et al., 2015). In most studies, SC are measured using diffusion tractography, which is based on the diffusion tensor reconstructed within WM. At the same time, FC is independently estimated using functional magnetic resonance imaging (fMRI) in GM (Chiang et al., 2015; Fotiadis et al., 2023; Liu et al., 2023; Zhao et al., 2021). However, because diffusion tractography-based SC and the statistical methods used for determining FC between different GM regions are fundamentally different, directly integrating these two approaches may provide limited insights into the SC underlying functional interactions (Avena-Koenigsberger et al., 2017). Excitingly, a study that combined DTI tractography and FC has shown a high overall correspondence between the major WM tracts in various regions and FC in WM. This demonstrates a close relationship between the BOLD signals in WM, its fiber tracts, and its anatomical connectivity (Wang et al., 2021). By integrating functional MRI signals from WM as a component in network analyses, we may obtain more comprehensive descriptions of brain function (Zu et al., 2024). Zhao et al. combined spatial-temporal correlations of functional signals with diffusion tensor orientations in the WM circuit from functional and diffusion magnetic resonance images (MRI). This approach helped uncover abnormal brain connectivity in Schizophrenia (Zhao et al., 2023). However, the complex interactions between structure and function in WM in patients with MDD remain largely unknown.

In this study, we proposed a novel measurement to characterize the structure-function relationship in WM tracts, known as SC-FC coupling, by quantifying the Euclidean distance between structural and functional signals in these tracts. The primary goals of the present study were to examine the alterations of SC-FC coupling in WM tracts in patients with MDD and to explore the associations between coupling and clinical symptoms. Furthermore, we assessed differences among three MDD subgroups: those with suicidal ideation (SI), those with a history of suicidal attempts (SA), and those non-suicidal (NS). This study aimed to provide a novel understanding of the neurobiological basis of MDD from a WM perspective.

## 2. Methods and materials

The data generation and analysis workflows are schematically represented in Fig. 1.

### 2.1. Participants

A total of 178 MDD patients were enrolled in the study, with 18 reporting SI, 24 with SA, and 51 NS. Additionally, 173 healthy controls (HCs) were recruited to match the patient group in key demographic variables. Details of inclusion/exclusion criteria and assessments are provided in the Supplementary materials.

This study was approved by the Ethics Committee of Yuhuangding Hospital and followed the ethical guidelines of the World Medical Association Declaration of Helsinki (Anon, 2013). All subjects provided informed consent.

### 2.2. Clinical assessments

Demographic information, including gender, age, BMI, and education level, was collected for all participants. Additionally, the duration of illness was recorded for patients diagnosed MDD. The severity of depressive symptoms, anxiety symptoms and cognitive were assessed using the 24-item Hamilton Depression Scale (HAMD) (Williams, 1988), the Hamilton Anxiety Rating Scale (HAMA) (Hamilton, 1959) and Montreal Cognitive Assessment Scale (MoCA) (Nasreddine et al., 2005), respectively. Furthermore, the risk of suicide in individuals with MDD was evaluated using the Nurses' Global Assessment of Suicide Risk (NGASR) (Cutcliffe and Barker, 2004). The total score on the NGASR scale indicates the level of suicide risk, with a higher score signifying a greater risk. Suicide ideation and suicidal attempts were defined by the relevant sub items of the NGASR scale. A summary of the demographic and clinical characteristics of all participants can be found in Table 1.

### 2.3. Image acquisition and preprocessing

Imaging data including T1-weighted images, resting-state fMRI, and DTI were obtained using a 3 T scanner (GE 750 W, GE Healthcare, the United States). Image acquisition parameters are detailed in the Supplementary materials. A routine T2-weighted image of axial brain was obtained for disease diagnosis.

Preprocessing of T1-weighted images involved bias correction, skull stripping, and normalization to the Montreal Neurological Institute standard space. DTI data preprocessing included denoising, B1-field correction, eddy current correction, DTI-T1 weighted alignment, and model fitting. Resting-state fMRI data preprocessing included slice-timing correction, motion correction, normalization, de-linear trending, temporal bandpass filtering, nuisance regression, and smoothing. Notably, WM and brain global signal were retained to preserve important neural signals. Further details on image preprocessing, tractography, and quality control can be found in the Supplementary materials.

2.4. Network construction

The ICBM-DTI-81 WM labels atlas was utilized to define network nodes, with 48 WM tract labels created through hand segmentation of diffusion MRI tensor maps from 81 subjects (Oishi et al., 2008). To construct the SC, a region descriptor was defined as a feature vector consisting of five diffusion metrics for each WM region. These metrics include FA, axial diffusivity (AD), radial diffusivity (RD), mean diffusivity (MD), and local diffusion homogeneity (LDH). For each participant, 5 diffusion metrics were z-normalized across regions to adjust for variations in value distributions between features. Spearman's correlation coefficients were then calculated based on the diffusion metrics between each paired WM region, forming a  $48 \times 48$  SC matrix. The FC matrix was estimated using Pearson's correlation coefficient between mean BOLD time series from all pairs of 48 WM tracts, and converted to z scores through Fisher's r-to-z transformation.

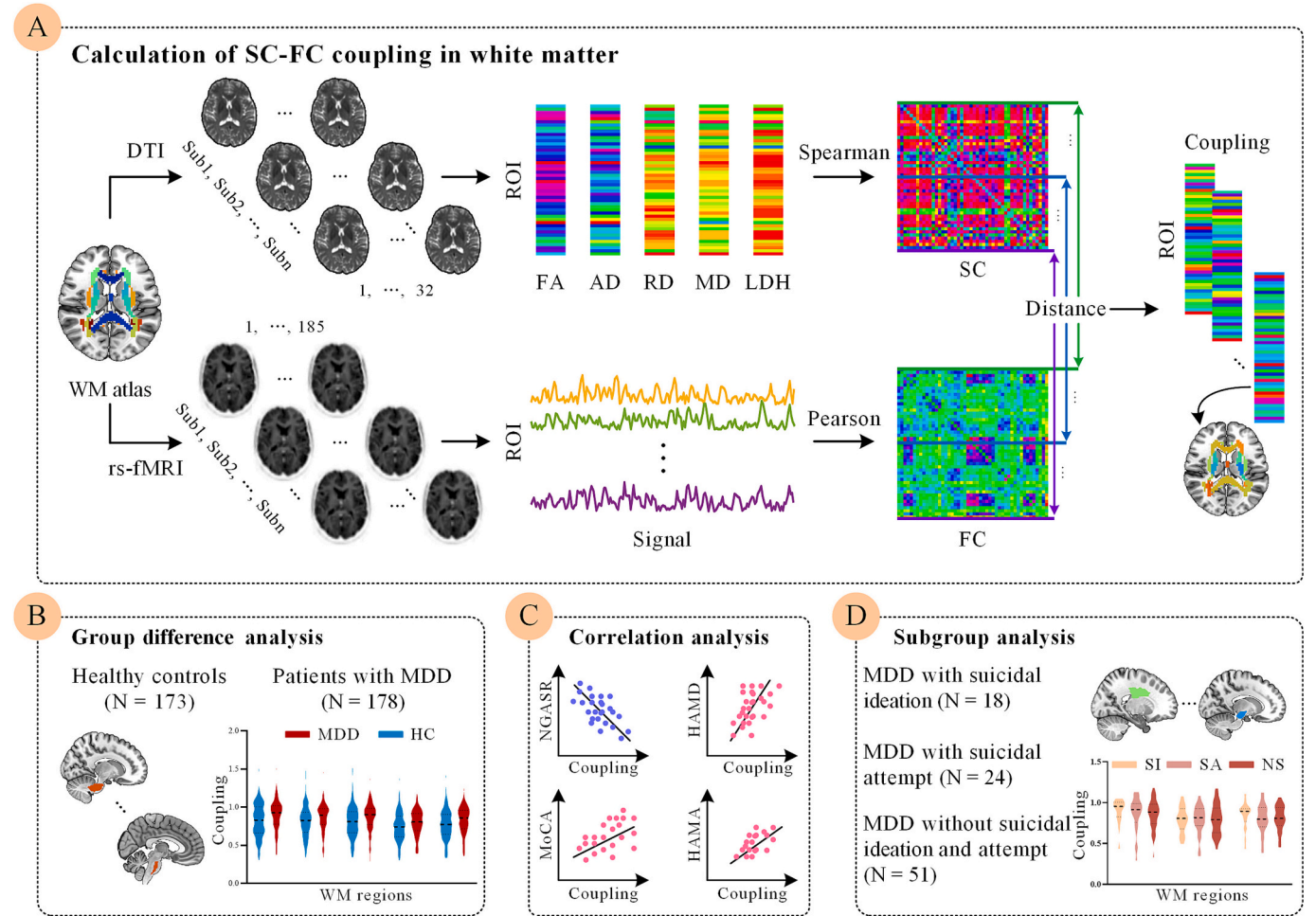
2.5. SC-FC coupling in WM

To evaluate the SC-FC coupling, the Euclidean distance between the corresponding rows of the SC and FC matrices was calculated for each subject. This process generated a 48-element vector, accurately capturing the coupling values within each WM tract for each participant.

**Table 1**  
Demographic and clinical characteristics of all subjects.

|                              | MDD<br>(n = 178) | HCS<br>(n = 173) | p value            |
|------------------------------|------------------|------------------|--------------------|
| Gender (male/female)         | 44/134           | 48/125           | 0.519 <sup>†</sup> |
| Age (year)                   | 27.796 ± 10.409  | 31.802 ± 11.359  | 0.383              |
| BMI                          | 22.443 ± 3.545   | 23.005 ± 3.789   | 0.321              |
| Education (year)             | 12.397 ± 3.366   | 16.100 ± 3.953   | 0.705              |
| Duration of illness (months) | 19.880 ± 25.451  | NA               | /                  |
| HAMD-24                      | 31.951 ± 8.086   | 3.317 ± 2.370    | <0.0001**          |
| NGASR                        | 8.441 ± 3.986    | NA               | /                  |
| MoCA                         | 23.521 ± 2.203   | 27.425 ± 1.317   | <0.0001**          |
| HAMA                         | 18.333 ± 5.383   | 1.614 ± 1.777    | <0.0001**          |

Notes: Group differences between the MDD and HCs groups were analyzed by two-sample t-test for age, BMI, education, HAMD-24, MoCA and HAMA or  $\chi^2$  (chi-squared) test for gender. Values are expressed as the mean ± SD. MDD, Major Depressive Disorder; HCs, Healthy Controls; BMI, Body Mass Index; HAMD-24, Hamilton Depression Scale-24; NGASR, Nurses' Global Assessment of Suicide Risk; MoCA, Montreal Cognitive Assessment Scale; HAMA, Hamilton Anxiety Scale; NA, Not Applicable; <sup>†</sup>, The  $\chi^2$  (chi-squared) test; \*,  $p < 0.05$ ; \*\*,  $p < 0.001$ .



**Fig. 1.** Data generation and analysis workflows. (A) The quantification of SC-FC coupling in WM tracts using the Euclidean distance between corresponding rows of the SC and FC matrices for each subject. (B) Group difference analysis comparing SC-FC coupling differences in each WM tract between MDD and HCs. (C) Correlation analysis between SC-FC coupling in WM tracts and clinical measures, i.e., NGASR, HAMD-24, MoCA, HAMA. (D) Difference in SC-FC coupling in WM tracts among three MDD subgroups. WM, white matter; SC, structural connectivity; FC, functional connectivity; MDD, Major depressive disorder; HCs, healthy controls; NGASR, Nurses' Global Assessment of Suicide Risk; HAMD-24, 24-item Hamilton Depression Rating Scale; MoCA, Montreal Cognitive Assessment Scale; HAMA, Hamilton Anxiety Rating Scale.

## 2.6. Statistical analysis

The demographic and clinical characteristics were analyzed using Statistical Product and Service Solutions (SPSS, version 26.0). Group differences between MDD and HCs were examined using a two-sample *t*-test for age, BMI, education, HAMD-24, MOCA, and HAMA, and a  $\chi^2$  (chi-squared) test for gender.

The Kolmogorov-Smirnov method was used for a normality test, revealing that the SC-FC coupling in most WM tracts did not have a normal distribution ( $p < 0.05$ ). Therefore, the non-parametric Mann-Whitney *U* test was performed to assess the differences in SC-FC coupling between groups, with a false discovery rate (FDR) correction applied ( $P_{FDR} < 0.05$ ). Then, we used a spearman correlation analysis to explore the association between clinical symptoms and SC-FC coupling. Furthermore, the Kruskal Wallis test was used to determine significant differences in SC-FC coupling across the three MDD subgroups. Post-hoc pairwise comparisons were carried out to explore the nature of these group differences, with a Bonferroni correction for multiple comparisons ( $P_{Bonferroni} < 0.05$ ).

## 2.7. Sensitivity analysis

We employed two additional distance algorithms, namely Cityblock and Chebychev, to further confirm the robustness of the SC-FC coupling. Initially, we calculated the intraclass correlation coefficients (ICC) values to assess the reliability of SC-FC coupling in WM tracts across different distance algorithms (Mcgraw and Wong, 1996). Subsequently, Bland-Altman plots were used to evaluate the consistency of case-control differences in SC-FC coupling, providing us with limits of agreement (LoA) for each measure. The LoA for the replication studies involving various distance algorithms were determined using a 95 % prediction interval for the differences in values.

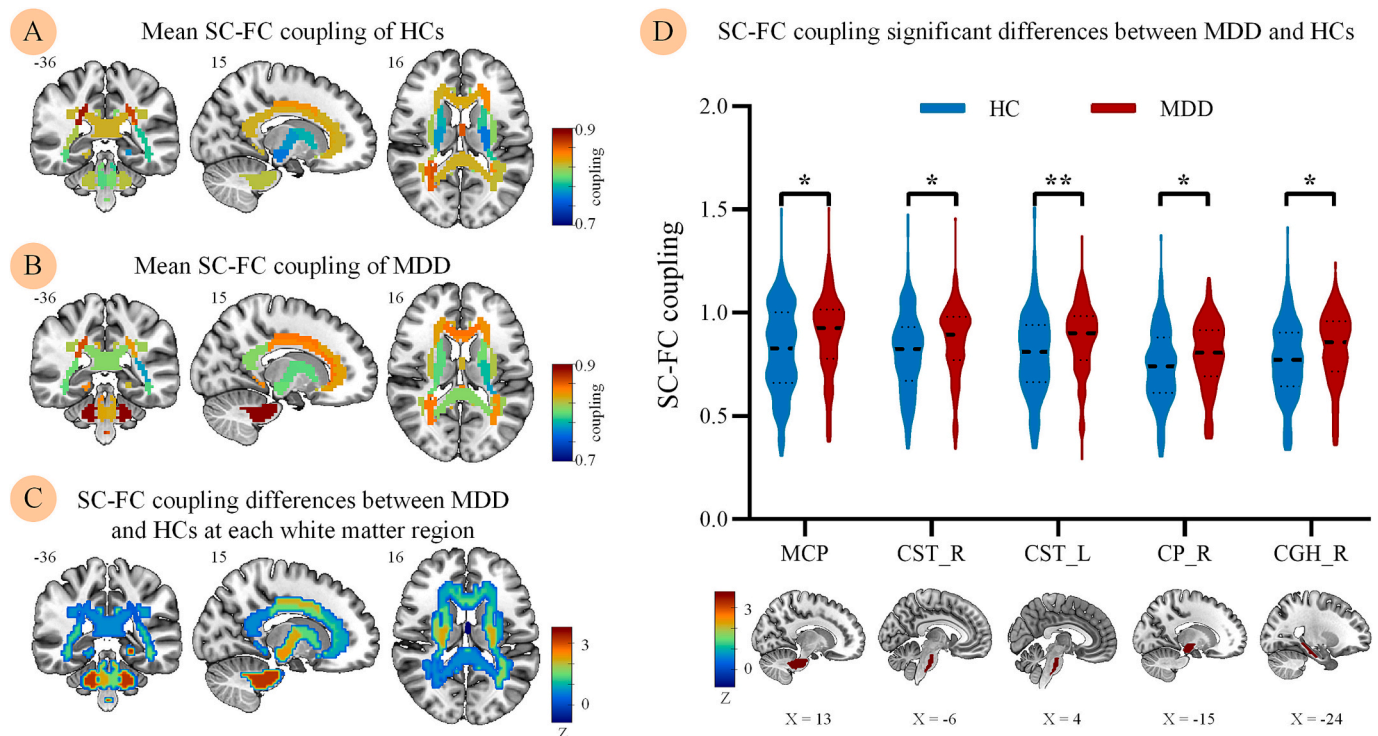
## 3. Results

### 3.1. Demographic and clinical characteristics

In total, 178 patients with MDD (134 females; age:  $27.796 \pm 10.409$  years; BMI:  $22.443 \pm 3.545$ ; education:  $12.397 \pm 3.366$  years; duration of illness:  $19.880 \pm 25.451$  months) and 173 HCs (125 females; age:  $31.802 \pm 11.359$  years; BMI:  $23.005 \pm 3.789$ ; education:  $16.100 \pm 3.953$ ) were included for analysis in this study. The demographic and clinical characteristics of these subjects are shown in Table 1. No significant difference was observed in the distributions of gender ( $p = 0.519$ ), age ( $p = 0.383$ ), BMI ( $p = 0.321$ ) and education ( $p = 0.705$ ) between MDD and HCs. Patients with MDD showed higher scores on the HAMD ( $p < 0.0001$ ) and HAMA ( $p < 0.0001$ ) scales, decreased cognitive function (MoCA,  $p < 0.0001$ ), and varying degrees of suicide risk (NGASR, mean = 8.441, SD = 3.986) compared with HCs.

### 3.2. Aberrant SC-FC coupling in WM tracts between patients with MDD and HCs

The average SC-FC coupling in WM tracts of the HCs group is shown in Fig. 2A, while the average SC-FC coupling in the MDD group is displayed in Fig. 2B. It was found that at the group level, SC-FC coupling was consistently positive (HCs: 0.740–0.885; MDD: 0.772–0.892) and varied considerably across the WM tracts. Although the spatial distributions of SC-FC coupling in MDD and HCs were similar, abnormal coupling was identified in certain regions (Fig. 2C). Patients with MDD exhibited altered SC-FC coupling in 5 WM tracts compared with HCs (Table 2, Fig. 2D). Specifically, increased SC-FC coupling were observed in the middle cerebellar peduncle (MCP,  $Z = 3.202$ ,  $P_{FDR} = 0.016$ ), bilateral corticospinal tract (right: CST\_R,  $Z = 3.137$ ,  $P_{FDR} = 0.016$ ; left: CST\_L,  $Z = 3.790$ ,  $P_{FDR} = 0.007$ ), right (CP\_R,  $Z = 3.219$ ,  $P_{FDR} = 0.016$ )



**Fig. 2.** Differences of SC-FC coupling in WM tracts between patients with MDD and HCs. (A) Average SC-FC coupling in WM tracts for the HCs group (range: 0.740–0.885). (B) Average SC-FC coupling in WM tracts for the MDD group (range: 0.772–0.892). (C) Case-control comparison (z-map) of SC-FC coupling in WM tracts. (D) Five WM tracts show statistically significant differences ( $P_{FDR} < 0.05$ ). In the violin plot, the dashed lines from top to bottom represent the first quartile, median, and third quartile, respectively. The bottom panel displays the inter-group difference z values for the five WM tracts. Compared with the HCs group, the MDD group shows significantly enhanced SC-FC coupling in MCP, bilateral CST, right CP and right CGH. MCP, middle cerebellar peduncle; CST, corticospinal tract; CP, cerebral peduncle; CGH, cingulum (hippocampus).

**Table 2**  
Altered SC-FC coupling in WM tracts in patients with MDD compared to HCs.

| Label | Abbreviation | $P_{FDR}$ | Z     | Region                       |
|-------|--------------|-----------|-------|------------------------------|
| 1     | MCP          | 0.016     | 3.202 | Middle cerebellar peduncle   |
| 7     | CST_R        | 0.016     | 3.137 | Right corticospinal tract    |
| 8     | CST_L        | 0.007     | 3.790 | Left corticospinal tract     |
| 15    | CP_R         | 0.016     | 3.219 | Right cerebral peduncle      |
| 37    | CGH_R        | 0.016     | 3.360 | Right cingulum (hippocampus) |

Notes: WM, white matter; MDD, Major Depressive Disorder; HCs, Healthy Controls.

and right cingulum (hippocampus) (CGH-R,  $Z = 3.360$ ,  $P_{FDR} = 0.016$ ). Importantly, no decreased in SC-FC coupling in WM tracts was observed in patients with MDD.

3.3. Correlation between SC-FC coupling and MDD symptoms

Upon examination of Fig. 3, it is evident that there were significant positive correlations between coupling strength in CP\_R and HAMA scores ( $r = 0.269$ ,  $P_{FDR} = 0.041$ ). Additionally, a significant negative correlation was identified between coupling strength in CGH-R and NGASR scores ( $r = -0.159$ ,  $P_{FDR} = 0.036$ ). However, no significant correlations were noted in HAMD and MoCA scores following FDR correction.

3.4. Disrupted SC-FC coupling in WM tracts among MDD subgroups

To investigate the changes in SC-FC coupling associated with different clinical characteristics, patients were grouped according to two behavioral symptoms: SI or SA. Specific alterations in SC-FC coupling were detected in MDD subgroup, including left anterior limb of internal capsule (AIC\_L), right external capsule (EC\_R), CST\_L, left superior cerebellar peduncle (SCP\_L) and right posterior limb of internal capsule (PIC\_R) (Table 3).

Compared with MDD with SA and NS, those with SI exhibited unique increases in SC-FC coupling in WM regions of AIC\_L ( $P_{Bonferroni} = 0.008$ ;  $P_{Bonferroni} = 0.005$ ), CST\_L ( $P_{Bonferroni} = 0.021$ ;  $P_{Bonferroni} = 0.019$ ), and decreases in EC\_R ( $P_{Bonferroni} = 0.023$ ;  $P_{Bonferroni} = 0.026$ ). Additionally, patients with SA showed a unique decrease in SC-FC coupling in SCP\_L compared to those with SI ( $P_{Bonferroni} = 0.028$ ), while MDD with SA exhibited a unique decrease in PIC\_R compared to NS ( $P_{Bonferroni} = 0.032$ ) (Fig. 4). Overall, MDD with SI showed higher SC-FC coupling in WM tracts compared to patients with SA and NS. This indicates that SC-

**Table 3**  
Alterations of SC-FC Coupling in WM tracts among MDD Subgroups.

| Group A vs. Group B | Abbreviation | Region                                   | $P_{Bonferroni}$ |
|---------------------|--------------|------------------------------------------|------------------|
| SI vs. SA           | AIC_L        | Left anterior limb of internal capsule   | 0.008            |
|                     | EC_R         | Right external capsule                   | 0.021            |
|                     | CST_R        | Right corticospinal tract                | 0.023            |
| SI vs. NS           | SCP_L        | Left superior cerebellar peduncle        | 0.028            |
|                     | AIC_L        | Left anterior limb of internal capsule   | 0.005            |
|                     | EC_R         | Right external capsule                   | 0.019            |
| SA vs. NS           | CST_R        | Right corticospinal tract                | 0.026            |
|                     | PIC_R        | Right posterior limb of internal capsule | 0.032            |

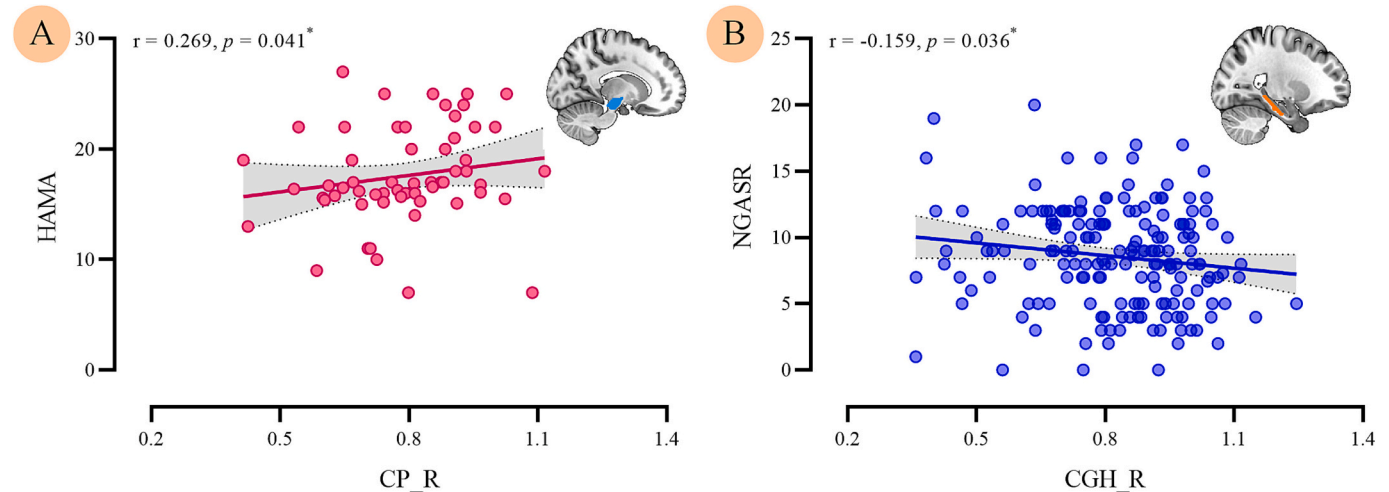
Notes: WM, white matter; SI, suicidal ideation; SA, suicidal attempts; NS, non-suicidal.

FC coupling in WM could be useful in distinguishing among different subgroups of MDD patients with various clinical characteristics.

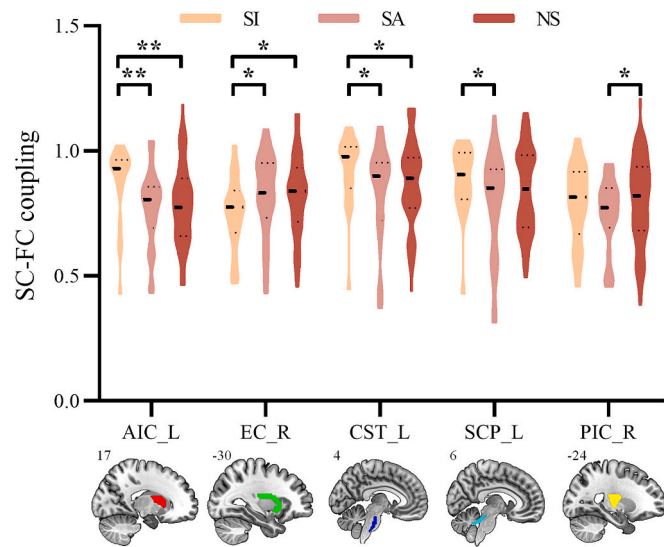
3.5. Sensitivity analysis

The ICC map results, comparing various distance algorithms, are illustrated in Fig. 5A. The mean ICC value for WM tracts comparing the Euclidean and Cityblock algorithms was 0.890 in HCs and 0.898 in patients with MDD, indicating high agreement. Similarly, the mean ICC value between the Euclidean and Chebychev algorithms was 0.977 in HCs and 0.968 in patients with MDD, showing strong reliability of SC-FC coupling across these algorithms. Upon closer examination of individual WM tracts, the ICCs for the Cityblock algorithm ranged from 0.812 to 0.975 in HCs and from 0.853 to 0.953 in patients with MDD. For the Chebychev algorithm, the ICCs ranged from 0.964 to 0.988 in HCs and from 0.960 to 0.980 in patients with MDD. These results indicate that despite variations in the robustness of SC-FC coupling across different WM tracts and between case-control groups, it consistently demonstrates high reliability.

In the Bland-Altman plot, there is good agreement in the case-control differences of SC-FC coupling values calculated with different distance algorithms within the same dataset. Specifically, the mean difference ( $\mu$ ) between the Euclidean and Cityblock algorithms was 0.0573, with LoA at  $\mu \pm 3.5927$ . For the Euclidean and Chebychev algorithms, the  $\mu$  was  $-0.0386$ , with LoA at  $\mu \pm 0.1668$  (Fig. 5B).



**Fig. 3.** Associations of SC-FC coupling in WM tracts and clinical symptoms. The significance level was set at  $P < 0.05$ , with FDR correction applied. The shaded area represents the 95 % confidence interval. FDR, false discovery rate; CP\_R, right cerebral peduncle; CGH-R, right cingulum (hippocampus); HAMA, Hamilton Anxiety Rating Scale; NGASR, Nurses' Global Assessment of Suicide Risk.



**Fig. 4.** Differences in SC-FC coupling in WM tracts among clinical subgroups. In the violin plot, the dashed lines from top to bottom represent the first quartile, median, and third quartile, respectively. The bottom panel displays the locations for WM tracts. \*\*, indicates  $p < 0.01$ ; \*, indicates  $p < 0.05$ ; AIC\_L, left anterior limb of internal capsule; EC\_R, right external capsule; CST\_L, left corticospinal tract; SCP\_L, left superior cerebellar peduncle; PIC\_R, right posterior limb of internal capsule.

#### 4. Discussion

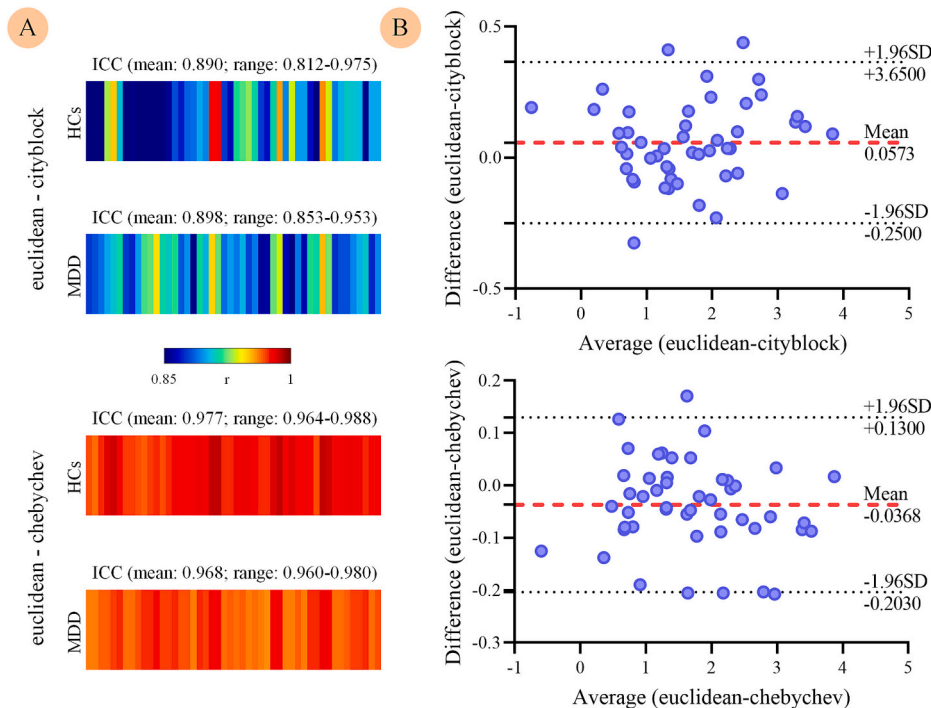
In this study, we proposed a novel measurement to characterize SC-FC coupling by quantifying the Euclidean distance between structural and functional signals in WM tracts. Our findings revealed aberrant SC-FC coupling in WM tracts between HCs and patients with MDD. Additionally, we identified associations between SC-FC coupling in WM tracts and clinical symptoms, as well as disrupted coupling among

different MDD subgroups.

##### 4.1. Aberrant development of SC-FC coupling in WM tracts in patients with MDD

At the group level, the average regional SC-FC coupling was entirely positive. These findings are consistent with previous studies in gray matter showing a positive correlation between SC and FC (Liao et al., 2024; Xu et al., 2024), supporting the notion that higher SC is generally associated with higher FC. The variability in SC-FC coupling across WM tracts underscores the heterogeneity of SC-FC relationships within the brain (Zhao et al., 2023).

The results highlighted the common increased SC-FC coupling in multiple WM tracts in patients with MDD. Mounting evidence has supported the MCP and CST tracts may influence alterations in motor control and somatosensory processing (Isa, 2012; Wang et al., 2023), which are relevant to the physical symptoms experienced by patients with MDD (Sacchet et al., 2014; Sanchez et al., 2023). Aberrant SC-FC Coupling in CP\_R may reveal potential abnormalities in the connections between the thalamus and cerebral cortex, which could affect sensory and emotional processing (Brown et al., 2017; Kang et al., 2018). Importantly, we found a significant positive correlation between coupling strength in CP\_R and HAMA scores, suggesting that increased SC-FC coupling in CP\_R may be associated with greater anxiety levels in patients with MDD. Meanwhile, previous research has shown that the CGH\_R is associated with episodic memory (Bubb et al., 2018), while clinical studies have revealed cingulum abnormalities in MDD (de Diego-Adeliño et al., 2014). Interestingly, the coupling strength in CGH\_R was found to be significantly negatively correlated with NGASR scores, suggesting that increased SC-FC coupling in CGH\_R may be associated with lower suicide risk in patients with MDD. Notably, no decreases in SC-FC coupling were observed in any WM tracts in patients with MDD. This suggests that alterations in SC-FC coupling in MDD may primarily manifest as increases rather than decreases, potentially reflecting compensatory mechanisms or underlying pathophysiological



**Fig. 5.** The results from the ICC map and Bland-Altman plot demonstrate the good reliability and reproducibility of SC-FC coupling in WM tracts. (A) The ICC map results, comparing various distance algorithms, show the consistency of the findings. (B) The Bland-Altman plot reveals good agreement in the case-control differences of SC-FC coupling values within the same dataset using different distance algorithms. ICC, intraclass correlation coefficient.

processes.

#### 4.2. Alterations of SC-FC coupling in WM among MDD subgroups

We found that MDD subgroups defined by the presence of SI or SA exhibited distinct patterns of SC-FC coupling alterations in several WM tracts. Specifically, the higher regional SC-FC coupling in WM observed in patients with SI compared to those with SA and NS may indicate underlying differences in neural circuitry associated with suicidal behavior. Convergent evidence has shown that reduced FA in AIC\_L implicates that the brain regions connected via AIC\_L could contribute to the impairment of the WM integrity (Dillon et al., 2018; Luttenbacher et al., 2022; Zhang et al., 2013), which may underlie the increased suicidal risk individuals with MDD. Further evidence for the involvement of the AIC is derived from deep brain stimulation (DBS) studies (Zhang et al., 2013), where bilateral stimulation of the AIC in MDD patients has been shown to induce acute improvements in mood (Cleary et al., 2015). Additionally, previous evidence has supported the increased MD of the CST\_L contributed to the elevated suicide score (Lei et al., 2024).

Additionally, a decreased SC-FC coupling in PIC\_R in patients with SA compared to NS may indicate changes in microstructure and functional signaling in WM. More specifically, the MDD group exhibited lower FA values in PIC\_R compared to HCs (Zhao et al., 2024). Furthermore, the regional homogeneity of WM showed an increase in PIC\_R in in patient with adolescent depression, potentially increasing the risk of suicide (Huang et al., 2024). Our findings suggest that the SC-FC decoupling in the PIC\_R could be a significant neurophysiological feature associated with depression-related suicidal behavior. Highlighting insights emerging from mapping intrinsic brain connectivity networks provide a potentially mechanistic framework for an understanding of aspects of human behavior, including suicide (Cheng et al., 2024; Mo et al., 2024; Zhang et al., 2024).

#### 4.3. Limitations

Several limitations should be noted. First, although the study included a relatively large multimodal neuroimaging data of MDD, the absence of subitems on the NGASR scale in some patients resulted in a small number of subgroups, particularly in the SI group. Larger studies with more participants are needed to confirm our results and to further explore the relationship between SC-FC coupling in WM tracts and clinical characteristics in MDD. Second, we matched the groups for gender, but more female participants were included in our study. A previous study showed that SC-FC coupling in gray matter is stronger in females than in males (Zhao et al., 2021). Future studies are needed to examine the effects of gender on SC-FC coupling in WM tracts. Last, the definition of SC-FC coupling in the present study provides a concise way to map the structure-function relationship, but it may sacrifice time-varying correspondence between structural and functional networks (Zamani Esfahlani et al., 2022). Future research can focus on how brain structure–function relationships change over time in patients with MDD.

#### 5. Conclusions

The identification of specific WM tracts with altered SC-FC coupling in patients with MDD offers new insights into the neural mechanisms underlying the disorder. Additionally, the SC-FC coupling in WM tracts showed specificity among different subgroups of patients with MDD. The unique changes in SC-FC coupling seen in WM tracts could potentially serve as a biomarker for categorizing depressed patients based on suicidality.

#### CRediT authorship contribution statement

**Tongpeng Chu:** Writing – original draft, Methodology,

Investigation, Funding acquisition, Formal analysis, Conceptualization. **Xiaopeng Si:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Xicheng Song:** Data curation, Conceptualization. **Kaili Che:** Methodology, Funding acquisition, Data curation. **Fanghui Dong:** Methodology, Formal analysis, Data curation. **Yuting Guo:** Methodology, Formal analysis, Data curation. **Danni Chen:** Methodology, Formal analysis, Data curation. **Wei Yao:** Formal analysis, Data curation. **Feng Zhao:** Methodology, Funding acquisition, Formal analysis, Conceptualization. **Haizhu Xie:** Data curation, Conceptualization. **Yinghong Shi:** Data curation, Conceptualization. **Heng Ma:** Writing – review & editing, Project administration, Investigation, Data curation, Conceptualization. **Dong Ming:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization. **Ning Mao:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

#### Funding

This study was supported by the Natural Science Foundation of Shandong Province (No. ZR2023QH084), National Natural Science Foundation of China (No. 62376191, No. 81925020, No. 82371933 and No. 62176140), Natural Science Foundation of Shandong Province (No. ZR2023QH107), Taishan Scholar Project of Shandong Province (NO. tsqn202103197) and Opening Project of Key Laboratory of Brain, Cognition and Education Sciences (South China Normal University), Ministry of Education.

#### Declaration of competing interest

All authors declare that they have no competing interests.

#### Acknowledgements

None.

#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2024.12.082>.

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